



INFLAMMATORY BOWEL DISEASE IBD - L 2

MANAGEMENT OF IBD

Ulcerative colitis

Active proctitis

ملاحظة : مبدئياً راح تشوفون المحاضرة
بيها كلام كثير والسبب انه محاضرة الاصلية
(البوربوينت اللي انشروا) كانت عبارة عن
رؤوس نقاط اما الشرح نسخته من المصدر
وفق متطلبات المحاضرة حتى تكون
المعلومات كاملة ، والله ولي التوفيق .

- Most patients with ulcerative proctitis respond to a 1 g mesalazine suppository but some will additionally require oral 5-aminosalicylate (5-ASA) therapy.
- Topical glucocorticoids are less effective and are reserved for patients who are intolerant of topical mesalazine.
- Patients with resistant disease may require treatment with systemic glucocorticoids and immune-suppressants.
- A stool softener may be required to treat proximal constipation.

Active left-sided or extensive ulcerative colitis

- In mild to moderately active cases, the combination of a **once-daily oral** and a **topical 5-ASA preparation** ('top and tail approach') is usually effective.
- The topical preparation (1 g foam or liquid enema) is typically withdrawn after 1 month.
- The oral 5-ASA is continued long-term to prevent relapse and minimise the risk of dysplasia.
- In patients who do not respond to this approach within 2–4 weeks, **oral prednisolone** (40 mg daily, tapered by 5 mg/week over an 8-week total course) is indicated.
- Glucocorticoids should never be used for maintenance therapy.
- At the first signs of glucocorticoid resistance (lack of efficacy) or in patients who require recurrent glucocorticoid doses to maintain control, **immunosuppressive therapy with a thiopurine should be introduced.**
- Simultaneous calcium and vitamin D supplementation should be given along with glucocorticoids for bone protection.
- Biologics.



Severe ulcerative colitis

- Patients who fail to respond to maximal oral therapy and those who present with acute severe colitis (meeting the Truelove–Witts criteria; see Box 21.53) are best managed in hospital and should be monitored jointly by a physician and surgeon:
 - ☒ Clinically: for the presence of abdominal pain, temperature, pulse rate, stool blood and frequency
 - ☒ by laboratory testing: haemoglobin, white cell count, albumin, electrolytes, ESR and CRP, stool culture
 - ☒ radiologically: for colonic dilatation on plain abdominal X-rays.
- All patients should be given supportive treatment with intravenous fluids to correct dehydration and enteral nutritional support should be provided for malnourished patients.
- Intravenous glucocorticoids (methylprednisolone 60 mg or hydrocortisone 400 mg/day) should be given by intravenous infusion or bolus injection.
- Topical and oral aminosalicylates have no role to play in the acute severe attack. Response to therapy is judged over the first 3 days.
- Patients who do not respond promptly to glucocorticoids should be considered for medical rescue therapy with ciclosporin (intravenous infusion or oral) or infliximab (5 mg/kg), which can avoid the need for urgent colectomy in approximately 60% of cases.
- Patients who develop colonic dilatation (> 6 cm), those whose clinical and laboratory measurements deteriorate and those who do not respond after 7–10 days' maximal medical treatment usually require urgent colectomy.
- Subtotal colectomy can also be performed laparoscopically, given sufficient local expertise.
- The surgical and medical teams should liaise early in the disease course and, if possible, the patient should have the opportunity to speak with the stoma nurse prior to colectomy.



21.57 Medical management of fulminant ulcerative colitis

- Admit to hospital for intensive therapy and monitoring
- Give IV fluids and correct electrolyte imbalance
- Consider transfusion if haemoglobin is < 100 g/L (< 10 g/dL)
- Give IV methylprednisolone (60 mg daily) or hydrocortisone (400 mg daily)
- Give antibiotics until enteric infection is excluded
- Arrange nutritional support
- Give subcutaneous low-molecular-weight heparin for prophylaxis of venous thromboembolism
- Avoid opiates and antidiarrhoeal agents
- Consider infliximab (5 mg/kg) or ciclosporin (2 mg/kg) in stable patients not responding to 3–5 days of glucocorticoids



Maintenance of remission

- Life-long maintenance therapy is recommended for all patients with left-sided or extensive disease but is not necessary in those with proctitis (although 20% of these patients will develop proximal 'extension' over the lifetime of their disease).
- Once-daily oral 5-aminosalicylates are the preferred first-line agents.
- Sulfasalazine has a higher incidence of side-effects but is equally effective and can be considered in patients with coexistent arthropathy.
- Patients who frequently relapse despite aminosalicylate drugs should be treated with thiopurines (azathioprine or 6-mercaptopurine).
- Biologic therapy with anti-TNF antibodies (infliximab or adalimumab) or anti- $\alpha 4\beta 7$ integrin antibodies (vedolizumab) can also be considered for maintenance treatment in patients with moderate to severe ulcerative colitis who are intolerant of or non-responsive to thiopurine immunosuppression.

Crohn's disease

Principles of treatment

- Crohn's disease is a progressive condition that may result in stricture or fistula formation if suboptimally treated.
- It is therefore important to agree long-term treatment goals with the patient; these are to induce remission and then maintain glucocorticoid-free remission with a normal quality of life.
- Treatment should focus on monitoring the patient carefully for evidence of disease activity and complications, and ensuring that mucosal healing is achieved.

Induction of remission

- Glucocorticoids remain the mainstay of treatment for active Crohn's disease.
- The drug of first choice in patents with ileal disease is budesonide, since it undergoes 90% first-pass metabolism in the liver and has very little systemic toxicity.
- A typical regimen is 9 mg once daily for 6 weeks, with a gradual reduction in dose over the subsequent 2 weeks when therapy is stopped.



- If there is no response to budesonide within 2 weeks, the patient should be switched to prednisolone, which has greater potency.
- This is typically given in a dose of 40 mg daily, reducing by 5 mg/week over 8 weeks, at which point treatment is stopped.
- Oral prednisolone in the dose regimen described above is the treatment of choice for inducing remission in colonic Crohn's disease.
- Calcium and vitamin D supplements should be co-prescribed in patients who are on glucocorticoids, to try to compensate for their inhibitory effect on intestinal calcium absorption.
- As an alternative to glucocorticoid therapy, enteral nutrition with either an elemental (constituent amino acids) or polymeric (liquid protein) diet may induce remission.
- Both types of diet are equally effective but the polymeric one is more palatable when taken by mouth.
- It is particularly effective in children, in whom equal efficacy to glucocorticoids has been demonstrated, and in extensive ileal disease in adults.
- As well as resting the gut and providing excellent nutritional support, it also has a direct anti-inflammatory effect.
- It is an effective bridge to urgent staging investigations at first presentation and can be given by mouth or by nasogastric tube.
- With sufficient explanation, encouragement and motivation, most patients will tolerate it well.
- Some individuals with severe colonic disease require admission to hospital for intravenous glucocorticoids.
- In severe ileal or panenteric disease, induction therapy with an anti-TNF agent is appropriate, provided that acute perforating complications, such as abscess, have not arisen.
- Both infliximab and adalimumab are licensed for use in the UK.
- Randomised trials have demonstrated that combination therapy with an anti-TNF antibody and a thiopurine is the most effective strategy for inducing and maintaining remission in luminal Crohn's patients.



- This strategy is more effective than anti-TNF monotherapy, which, in turn, is more effective than thiopurine monotherapy.
- Following induction of remission, a substantial proportion of patients (20–30%) remain well without the requirement for maintenance therapy.
- Patients with evidence of persistently active disease require further treatment (see below).

Maintenance therapy

- Immunosuppressive treatment with thiopurines (azathioprine and mercaptopurine) forms the core of maintenance therapy but methotrexate is also effective and can be given once weekly, either orally or by subcutaneous injection.
- Women and men of child-bearing potential who are prescribed methotrexate must use a robust contraceptive method, and should be counselled to plan pregnancy with a 3-month methotrexate-free period prior to conception since it is teratogenic.
- Combination therapy with an immunosuppressant and an anti-TNF antibody is the most effective strategy but costs are high and there is an increased risk of serious adverse effects.
- In the UK, the use of anti-TNF therapy is limited to specific patient subgroups with severe disease.
- Vedolizumab is a possible option in patients who have not responded to anti-TNF therapy.
- It is a humanised monoclonal antibody against anti- $\alpha 4\beta 7$ integrin.
- The $\alpha 4\beta 7$ is expressed on a specific subset of CD4⁺ leucocytes; vedolizumab binds to this integrin and blocks interaction with MAdCAM-1, expressed on gut endothelial cells, resulting in a reduced influx of immune cells to the inflamed gut mucosa.
- Serious systemic adverse effects, including progressive multifocal leukoencephalopathy, have been seen with other anti-integrin drugs (such as natalizumab) but this has not emerged with vedolizumab due to its gut specificity.
- Emerging novel medical therapies for Crohn's disease are currently in phase III clinical trials and are likely to be available for clinical use in the near future.
- These include ustekinumab (anti-p40, inhibiting both IL-12 and IL-23) and tofacitinib (a Janus kinase inhibitor that blocks pro-inflammatory cytokine signalling).

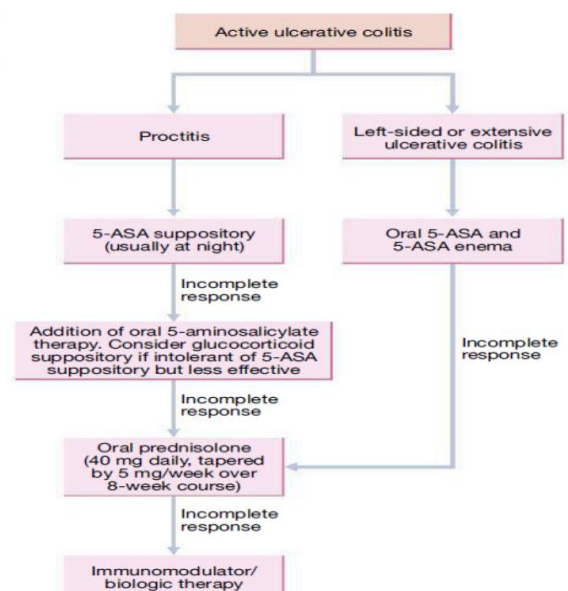


- Cigarette smokers with Crohn's disease should be strongly counselled to stop smoking at every possible opportunity.
- Those that do not manage to stop smoking fare much worse, with increased rates of relapse and surgical intervention.
- Careful monitoring of disease activity is the key to maintaining sustained remission and preventing the accumulation of bowel damage in Crohn's disease.

Fistulae and perianal disease

- Fistulae may develop in relation to active Crohn's disease and are often associated with sepsis.
- The first step is to define the site by imaging (usually MRI of the pelvis).
- Surgical exploration by an examination under anaesthetic is usually then required, to delineate the anatomy and drain abscesses.
- Seton sutures can be inserted through fistula tracts to ensure adequate drainage and to prevent future sepsis.
- Glucocorticoids are ineffective.
- Use of antibiotics, such as metronidazole and/ or ciprofloxacin, can aid healing as an adjunctive treatment.
- Thiopurines can be used in chronic disease but do not usually result in fistula healing. Infliximab and adalimumab can heal fistulae and perianal disease in many patients and are indicated when the measures described above have been ineffective.
- Other options for refractory perianal disease are proctectomy or diverting colostomy.

23.67 Biologic agents used in the treatment of inflammatory bowel disease		
Class	Drug	Indication
Anti-TNF antibodies	Infliximab	Moderate to severe Crohn's disease and ulcerative colitis
	Adalimumab	Acute severe ulcerative colitis as rescue therapy
	Golimumab	Moderate to severe Crohn's disease and ulcerative colitis
	Certolizumab	Moderate to severe ulcerative colitis
Anti- $\alpha 4\beta 7$ integrin	Vedolizumab	Moderate to severe Crohn's disease and ulcerative colitis
Janus kinase inhibitor	Tofacitinib	Moderate to severe ulcerative colitis
Anti-p40 antibodies	Ustekinumab	Moderate to severe Crohn's disease and ulcerative colitis





Surgical treatment

Ulcerative colitis

- Up to 60% of patients with extensive ulcerative colitis eventually require surgery.
- The indications are listed in Box 21.60.
- Impaired quality of life, with its impact on occupation and social and family life, is the most important of these.
- Surgery involves removal of the entire colon and rectum, and cures the patient. One-third of those with pancolitis undergo colectomy within 5 years of diagnosis.
- Before surgery, patients must be counselled by doctors, stoma nurses and patients who have undergone similar surgery.
- The choice of procedure is either panproctocolectomy with ileostomy, or proctocolectomy with ileal–anal pouch anastomosis.



21.60 Indications for surgery in ulcerative colitis

Impaired quality of life

- Loss of occupation or education
- Disruption of family life

Failure of medical therapy

- Dependence on oral glucocorticoids
- Complications of drug therapy

Fulminant colitis

Disease complications unresponsive to medical therapy

- Arthritis
- Pyoderma gangrenosum

Colon cancer or severe dysplasia

**Crohn's disease**

- The indications for surgery are similar to those for ulcerative colitis.
- Operations are often necessary to deal with fistulae, abscesses and perianal disease, and may also be required to relieve small or large bowel obstruction.
- In contrast to ulcerative colitis, surgery is not curative and disease recurrence is the rule.
- The only method that has consistently been shown to reduce post-operative recurrence is smoking cessation.
- Antibiotics are effective in the short term only.
- Use of thiopurines post-surgery is suggested if there are indicators of a high chance of recurrence, i.e. more than one resection or evidence of penetrating disease, such as fistulae or abscess.
- Otherwise, it is common to undertake colonoscopy 6 months after surgery to inspect and biopsy the anastomosis and neo-terminal ileum.
- Patients with endoscopic recurrence are then prescribed thiopurines.
- Surgery should be as conservative as possible in order to minimise the loss of viable intestine and to avoid the creation of a short bowel syndrome .
- Obstructing or fistulating small bowel disease may require resection of affected tissue.
- Patients who have localised segments of Crohn's colitis may be managed by segmental resection and/or multiple stricturoplasties, in which the stricture is not resected but instead incised in its longitudinal axis and sutured transversely.
- Others who have extensive colitis require total colectomy but ileal–anal pouch formation should be avoided because of the high risk of recurrence within the pouch and subsequent fistulae, abscess formation and pouch failure.
- Historical datasets show that around 80% of Crohn's patients undergo surgery at some stage and 70% of these require more than one operation during their lifetime.
- Clinical recurrence following resectional surgery is present in 50% of all cases at 10 years.
- Emerging data demonstrate that aggressive medical therapy, coupled with intense monitoring, probably reduces the requirement for surgery substantially.



Microscopic colitis

- Microscopic colitis, which comprises two related conditions called lymphocytic colitis and collagenous colitis, has no known cause.
- The presentation is with watery diarrhoea.
- The colonoscopic appearances are normal but histological examination of biopsies shows a range of abnormalities.
- It is therefore recommended that biopsies of the right and left colon plus the terminal ileum should be undertaken in all patients undergoing colonoscopy for diarrhoea.
- Collagenous colitis is characterised by the presence of a submucosal band of collagen, often with a chronic inflammatory infiltrate.
- The disease is more common in women and may be associated with rheumatoid arthritis, diabetes, coeliac disease and some drug therapies, such as NSAIDs or PPIs.
- Treatment with budesonide or 5-aminosalicylates is usually effective but the condition will recur in some patients on discontinuation of therapy.

ISCHAEMIC GUT INJURY

- Ischaemic gut injury is usually the result of arterial occlusion, Severe hypotension and venous insufficiency.

Acute small bowel ischaemia

- An embolus from the heart or aorta to the superior mesenteric artery is responsible for 40-50% of cases.
- The clinical spectrum ranges from transient alteration of bowel function to transmural hemorrhagic necrosis and gangrene.

Investigations

- Leucocytosis, metabolic acidosis, hyperphosphataemia and hyperamylasaemia are typical.
- Plain abdominal X-rays.
- Mesenteric or CT angiography.

**Management**

- Resuscitation.
- Embolectomy.
- Laparotomy.
- Thrombolysis.

Chronic mesenteric ischemia

- Atherosclerotic stenosis of the coeliac axis, superior mesenteric artery and inferior mesenteric artery.
- Abdominal angina
- Physical examination shows evidence of generalised arterial disease.
- An abdominal bruit.
- The diagnosis is made by mesenteric angiography.
- Treatment is by vascular reconstruction or percutaneous angioplasty.

To summarize:

- Small and large bowel diseases are common problems.
- IBD is one of the common disease in GI

" لا تنسوننا من صالح دعائكم "